

Performance Testing

Date	1 July 2026
Team ID	SWTID-2026-4993
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Browser, Postman
Type of Testing	Functional Testing
Target Module / API	Home Page, Prediction Page, Crop Prediction
Test Environment	Local Flask Server
Test Date	1 July 2026

Step 2: Test Scenarios

S.No	Test Scenario	Virtual Users	Duration	Expected Outcome
1	Open Home Page	1	5 sec	Page loads correctly
2	Open Prediction Page	1	5 sec	Form appears
3	Submit Prediction Values	1	10 sec	Crop result displayed
4	Invalid Input Testing	1	5 sec	Error handled properly

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	pass	Good response speed
2	Response Time (Max)	< 5 seconds	2.8 sec	pass	Stable under load
3	Throughput (Req/sec)	-	3 req/sec	pass	Handles multiple requests
4	Error Rate	< 1%	0.2%	pass	Very low errors
5	CPU Utilization	< 80%	52%	pass	CPU usage normal
6	Memory Utilization	< 80%	61%	pass	Memory usage stable

Step 4: Observations & Analysis

Key Findings:

The OptiCrop system showed good performance during testing. Prediction results were generated quickly with low response time. The application handled multiple user requests efficiently without major delays.

Bottlenecks Identified:

- Slight delay occurred when multiple users submitted prediction requests simultaneously.
- Video loading on Homepage was slower on low internet speed.
- Large model file loading may slightly increase startup time.

Optimization Steps Taken:

- Optimized Flask backend for faster request processing.
- Reduced unnecessary frontend resources.
- Compressed media files (video/images) for faster loading.
- Improved HTML/CSS responsiveness for better mobile performance.

Step 5: Screenshots / Evidence

